PROGRAM 2 — EMAIL VALIDATION SYSTEM

3.1 Introduction and Background

Email validation is an essential part of many software applications. This program develops a console-based Email Validation System in Python that checks an email address using structural rules and provides clear validation feedback. It applies character-level string analysis using Python's built-in string methods without regular expressions.

3.2 Objectives

- Design and implement a console-based Email Validation System.
- Check for a single '@' symbol and absence of spaces.
- Validate the local part and domain separately.
- Verify the domain format and top-level domain (TLD).
- Accept only predefined common TLDs.
- Test the validator with valid and invalid email addresses.

3.3 Problem Statement / Driving Question

How can an email address be validated using simple rule-based checks while providing clear and specific error messages for invalid formats?

3.4 Methodology

Program Structure: Reused the box-drawing helper functions to create a consistent bordered menu and result interface with an ASCII-art banner.

Local Part and Domain Validation: Developed name_check() to validate the local part and domain_check() to verify the domain structure and TLD.

Overall Email Validation Logic: Implemented email_checker() to perform basic format checks, validate the local and domain parts, and verify the TLD against a predefined list.

Menu-Driven Control Flow: Built a continuous menu with **Check Email** and **Exit** options, and tested the validator with different valid and invalid email formats.

3.5 Expected Outcomes / Deliverables

- A functional Email Validation System.
- Clear validation messages for invalid email formats.
- A user-friendly boxed console interface.
- Practical understanding of rule-based string validation.
- Scope for future improvements such as regex support and an expandable TLD list.